

SOURCE OF DATASET

PASCAL VOC 2012 DATASET



DATA PREPROCESSING

Data Augmentation

- 1) Random Resizing and Cropping
- 2) Flip and Rotation
- 3) Normalize pixel values

Splitting Dataset

- 1) Train/Validation/Test Split
- 2) Ensure balanced classes: Ensure that each class is well represented in each split

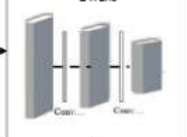
Handling Imbalanced Data

Class balancing :-using techniques like oversampling or data augmentation for minority classes to balance the dataset

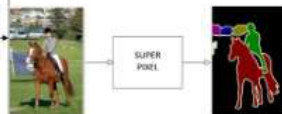


INPUT IMAGES

CONVOLUTION LAYERS



SALENCY DETECTION

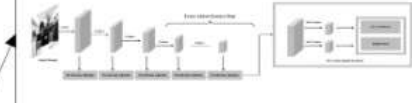


PROPOSAL GENERATION

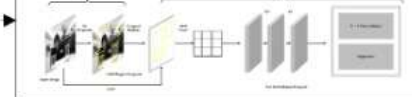


PROPOSED MODELS

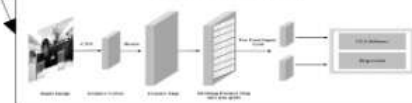
SINGLE SHOT DETECTOR



FAST RCNN



YOU ONLY LOOK ONCE (YOLO)



EVALUATION OF MODELS

Predicted class	
P	N
True Positives (TP)	False Negatives (FN)
False Positives (FP)	True Negatives (TN)
Confusion Matrix	



DETECTION AND RECOGNITION